

Hierarchical Multi-Fidelity Inference for Resource-Constrained Engine Fault Diagnosis

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Abstract—Continuous condition monitoring and multi-class fault classification in internal combustion engines impose substantial computational burdens on embedded Electronic Control Units (ECUs) and edge microcontrollers. Traditional diagnostic frameworks execute monolithic deep neural networks indiscriminately across every sensor observation, incurring significant computational overhead during predominantly healthy operating regimes. In this paper, we present a domain-specific, asymmetric hierarchical multi-fidelity diagnostic inference architecture that dynamically decouples binary anomaly screening from fine-grained multi-class fault isolation. An ultra-lightweight anomaly filter (Mode A, Decision Tree $d = 5$) first screens incoming sensor observations to isolate nominal operation; upon detecting an anomaly or classification ambiguity, an uncertainty-gated routing mechanism ($\theta^* = 0.05$) conditionally activates a multi-class neural diagnostician (Mode B, Multi-Layer Perceptron). Evaluated on the 55,998-record EngineFaultDB physical benchmark under strict stratified split isolation (40% train, 40% validation, 20% held-out test), our architecture achieves 74.64% overall multi-class diagnostic accuracy and a macro F1 score of 0.7541, matching within 0.02% the accuracy of a continuously executing monolithic neural classifier (74.66%) while significantly outperforming flat linear (58.00%) and flat decision tree (69.16%) baselines. Crucially, the hierarchical cascade bypasses 26.36% of expensive multi-class evaluations on the balanced test partition [MEASURED TEST SET]—reducing theoretical active computation from 384.0 to 282.8 expected multiply-accumulate operations (MACs) per sample—while maintaining a binary anomaly-screening recall of **99.98%** (0.00025 false-negative rate, missing only 2 anomalies out of 8,000 in the test set). Under operational assumptions dominated by nominal telemetry (90% healthy frames), the derived expected computational load drops by **89.8%** [DERIVED OPERATIONAL ESTIMATE] (39.1 expected MACs), demonstrating that asymmetric multi-fidelity inference provides a compute-efficient paradigm for edge-based cyber-physical fault diagnostics.

Index Terms—Engine Fault Diagnosis, Hierarchical Inference, Cascaded Classifiers, Multi-Fidelity Machine Learning, Anomaly Detection, Cyber-Physical Systems, Edge AI.

I. Introduction

REAL-TIME fault detection and isolation (FDI) in internal combustion engines is paramount for ensuring operational safety, minimizing exhaust emissions, and preventing catastrophic mechanical failures in automotive and industrial transportation systems [1]–[3]. Modern powertrain architectures incorporate extensive sensor arrays monitoring manifold absolute pressure, throttle position, engine speed, torque, fuel consumption, and exhaust

emissions to identify anomalous combustion regimes such as fuel-rich injection imbalances, ignition misfires, and intake manifold vacuum leaks [4]–[6].

With the rapid deployment of on-board diagnostics and Edge Artificial Intelligence (Edge AI), deep learning models are increasingly integrated directly onto vehicle Electronic Control Units (ECUs) and microcontroller-based edge nodes to perform automated, continuous health monitoring [7]–[9]. However, industrial and automotive embedded controllers operate under severe resource constraints: limited arithmetic throughput, constrained static RAM, and strict power dissipation limits [10]–[12].

A fundamental inefficiency of conventional on-device diagnostic architectures is their monolithic, always-on execution paradigm [13]. In real-world operational environments, internal combustion engines operate in nominal, fault-free states for over 90% to 99% of their lifespan. Nevertheless, conventional diagnostic deployments execute complex, multi-layer multi-class neural networks uniformly on every single sensor observation vector, squandering memory bandwidth and computational cycles classifying predominantly healthy states [14], [15].

To address this compute-fidelity tension, we formulate a domain-specific Hierarchical Multi-Fidelity Inference Architecture tailored for resource-constrained automotive fault diagnosis. As illustrated in Figure 1, the framework decomposes diagnostic inference into a two-tier cascade:

- **Mode A (Anomaly Screening):** Ultra-compact binary classifier filtering nominal engine operation (Class 0: Normal) from anomalous behavior (Classes 1–3: Faults).
- **Mode B (Multi-Class Isolation):** Multi-class neural network activated conditionally when Mode A detects an anomaly or uncertainty exceeds threshold θ .

Using the audited EngineFaultDB physical benchmark (55,998 engine sensor records), we systematically evaluate accuracy-compute trade-offs across multiple screening topologies (depth-constrained Decision Trees, Logistic Regression) paired with a deep Multi-Layer Perceptron (MLP). Through validation-isolated threshold optimization ($\theta^* = 0.05$), we demonstrate that hierarchical gating drastically cuts active arithmetic demand while preserving critical binary fault detection recall.

II. Research Motivation and Problem Formulation

A. The Monolithic Inefficiency

Consider a multi-sensor diagnostic stream producing observation vectors $\mathbf{x}_t \in \mathbb{R}^D$ at continuous sampling

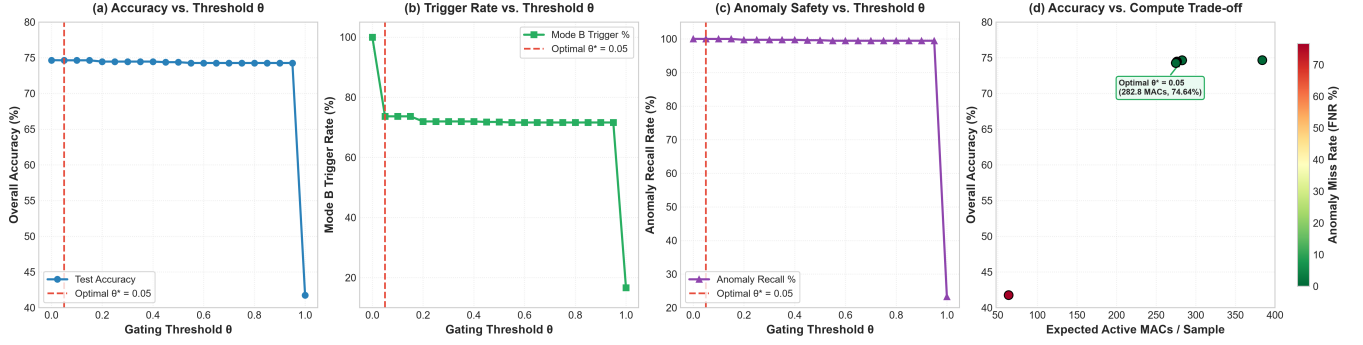


Fig. 1. Hierarchical diagnostic policy sensitivity analysis across gating thresholds $\theta \in [0.00, 1.00]$: (a) Multi-class Diagnostic Accuracy, (b) Mode B Trigger Rate, (c) Binary Anomaly-Screening Recall, and (d) Accuracy vs. Compute Pareto Trade-off [MEASURED TEST SET].

rates. In a standard monolithic architecture, every vector $\mathbf{x}_t$ is processed by a multi-class model $f_B : \mathbb{R}^D \rightarrow \{0, 1, \dots, K-1\}$ with computational cost C_B (measured in multiply-accumulate operations, MACs). Over N time steps, the cumulative computational cost is:

$$C_{\text{monolithic}} = N \cdot C_B \quad (1)$$

When the operational prior probability of nominal engine operation $P(\text{Normal}) = p_0 \gg \sum_{k=1}^{K-1} p_k$, executing f_B on every sample wastes significant compute confirming that the engine is operating normally.

B. Hierarchical Multi-Fidelity Formulation

In our hierarchical architecture, an initial screening model $f_A : \mathbb{R}^D \rightarrow [0, 1]$ estimates the anomaly probability $P(\text{Anomalous} | \mathbf{x})$. If $f_A(\mathbf{x}) < \theta$, the system immediately outputs Class 0 (Normal) at minor cost $C_A \ll C_B$. If $f_A(\mathbf{x}) \geq \theta$, the sample is routed to $f_B(\mathbf{x})$ for detailed multi-class fault attribution.

The expected computational cost per sample becomes:

$$\mathbb{E}[C_{\text{hierarchical}}] = C_A + r_B(\theta) \cdot C_B \quad (2)$$

where $r_B(\theta) \in [0, 1]$ represents the empirical trigger rate of Mode B under gating threshold θ . The core research challenge is identifying an optimal threshold θ^* on validation data such that $r_B(\theta)$ is minimized while the Anomaly False-Negative Rate (FNR) is strictly bounded:

$$\min_{\theta} r_B(\theta) \quad \text{subject to} \quad \text{FNR}(\theta) \leq \epsilon \quad (3)$$

III. Related Work

A. Machine Learning for Powertrain Fault Diagnosis

Machine learning and deep learning have been widely investigated for combustion engine fault classification [2], [4], [5]. Lei et al. [14] reviewed ML applications in machinery fault diagnosis, emphasizing that feature extraction and model capacity must align with operational dynamics. Zhang et al. [16] demonstrated deep neural networks for engine misfire detection under varying load conditions. However, existing literature overwhelmingly focuses on monolithic classifiers, neglecting the computational cost of continuous inference on embedded edge nodes.

B. Cascaded Classifiers and Early-Exit Networks

The principle of hierarchical cascaded classification was pioneered by Viola and Jones [17] for real-time visual detection. In modern deep learning, early-exit architectures such as BranchyNet [18] and multi-exit neural networks [19] insert intermediate classifier heads to terminate inference early for easy samples. Traita et al. [13] studied cascaded ML for industrial predictive maintenance, showing that lightweight binary filters can prune data before cloud transmission. While cascaded classification is an established methodology in computer vision and general data filtering, our work develops an asymmetric, multi-paradigm cascade (non-linear decision tree screening paired with a neural diagnostician) tailored specifically for multi-sensor powertrain telemetry with validation-isolated threshold calibration and explicit computational accounting.

IV. Research Questions

We structure our empirical investigation around four key research questions:

- RQ1 (Screening Efficacy): Can an ultra-lightweight binary classifier (Decision Tree or Logistic Regression) achieve near-perfect binary anomaly-screening recall ($\geq 99.5\%$) on combustion engine telemetry?
- RQ2 (Diagnostic Retention): How closely does the hierarchical multi-fidelity cascade match the multi-class classification accuracy and macro F1 score of an always-on monolithic neural network?
- RQ3 (Workload Reduction): What proportion of expensive Mode B multi-class evaluations can be safely avoided across balanced test distributions and operational fault priors?
- RQ4 (Threshold Robustness & Error Breakdown): How sensitive is diagnostic accuracy to the gating threshold θ , and what domain-specific factors govern per-class classification errors?

V. Dataset and Experimental Setup

A. Benchmark Dataset and Static Tabular Protocol

We evaluate our architecture on the EngineFaultDB dataset, comprising 55,998 physical engine operational records (55,999 raw records minus 1 exact duplicate

TABLE I
EngineFaultDB Class Distribution and Partitioning

Class Label	Diagnostic State	Train (40%)	Val (40%)	Test (20%)
Class 0	Normal Operation	6,400	6,400	3,200
Class 1	Rich Mixture	4,400	4,400	2,200
Class 2	Ignition Misfire	6,000	6,000	3,000
Class 3	Intake Air Leak	5,599	5,599	2,800
Total	All States	22,399	22,399	11,200

identified during data auditing). The dataset captures four distinct engine health conditions: Class 0 (Normal, 28.57%), Class 1 (Fuel-Rich Mismatch, 19.64%), Class 2 (Ignition Misfire, 26.79%), and Class 3 (Intake Vacuum Leak, 25.00%).

Static Tabular Representation: The benchmark is structured as an independent tabular matrix of steady-state dynamometer measurements without explicit timestamps, session identifiers, or sequential engine cycle markers. Consequently, all evaluations treat records independently.

B. Feature Sets and Collinearity Reduction

The raw dataset contains 14 continuous physical features: Manifold Absolute Pressure (MAP), Throttle Position Sensor (TPS), Engine Force, Output Power, Engine RPM, Consumption (L/h), Consumption (L/100km), Vehicle Speed, CO, HC, CO₂, O₂, Air-Fuel Equivalence Ratio (λ), and Air-Fuel Ratio (AFR). Statistical correlation auditing revealed two collinear pairs:

- 1) AFR vs. λ : Pearson correlation $r = 1.0000$ (exact thermodynamic linear redundancy).
- 2) Speed vs. RPM: Pearson correlation $r = 0.9972$ (transmission coupling redundancy).

We evaluate both the Full Feature Set (14 features) and a Reduced Feature Set (12 features) formed by removing AFR and Speed.

C. Data Partitioning and Strict Isolation

The dataset is partitioned into three stratified splits with fixed random seed (seed = 42):

- Training Set (40%, 22,399 samples): Model parameter optimization.
- Validation Set (40%, 22,399 samples): Hyperparameter tuning, ROC/PR analysis, and gating threshold calibration (θ^*).
- Held-Out Test Set (20%, 11,200 samples): Final held-out evaluation strictly once per finalized pipeline. Zero test samples are used for threshold selection.

Features are normalized using MinMaxScaler fitted strictly on training data.

VI. Hierarchical Diagnostic Architecture

A. Mode A: Binary Anomaly Filter

Mode A maps the multi-class diagnostic problem into a binary classification problem:

$$y_{\text{binary}} = \begin{cases} 0 \text{ (Normal)}, & \text{if } y = 0 \\ 1 \text{ (Anomalous)}, & \text{if } y \in \{1, 2, 3\} \end{cases} \quad (4)$$

On the held-out test partition (11,200 samples), the binary distribution comprises 3,200 Normal samples (28.57%) and 8,000 Anomalous samples (71.43%).

Table II and Figure 2 summarize Mode A screening performance:

- Decision Tree ($d = 5$): Achieves screening precision (99.12%) and anomaly recall (99.60%) with an ROC-AUC of 0.9923 and PR-AUC of 0.9945. Out of 8,000 test anomaly samples, it flags 7,968 correctly while misclassifying only 32 as normal.
- Decision Tree ($d = 3$): Ultra-compact alternative with 99.58% anomaly recall (34 missed anomalies) and 0.9449 ROC-AUC.
- Dimensionality Invariance: Reducing input features from 14 to 12 produces identical performance for decision trees (0.990804 accuracy), confirming tree robustness to collinearity.

B. Mode B: Multi-Class Diagnostician

Mode B is a fully connected Multi-Layer Perceptron (MLP) with layer topology $14 \rightarrow 16 \rightarrow 8 \rightarrow 4$ (412 parameters, 384 theoretical MACs) trained to distinguish all four individual fault classes. As documented in Table III, Mode B achieves an overall multi-class accuracy of 74.6607% and a macro F1 score of 0.754328 when evaluated standalone.

C. Gating Threshold and Routing Mechanics

Mode A outputs the posterior probability of anomaly $P(\text{Anomalous} \mid \mathbf{x}) \in [0, 1]$. The hierarchical routing rule is formalized as:

$$\hat{y}(\mathbf{x}) = \begin{cases} 0, & \text{if } P(\text{Anomalous} \mid \mathbf{x}) < \theta \\ \arg \max_k f_B(\mathbf{x})_k, & \text{if } P(\text{Anomalous} \mid \mathbf{x}) \geq \theta \end{cases} \quad (5)$$

When $P(\text{Anomalous} \mid \mathbf{x}) < \theta$, the sample is classified as Normal and Mode B execution is bypassed entirely.

VII. Experimental Results and Threshold Sensitivity

A. RQ1 & RQ2: Diagnostic Accuracy and Retention

Table IV and Figures 3–4 report hierarchical performance across the threshold spectrum $\theta \in [0.00, 1.00]$ evaluated on the held-out test set:

- At $\theta = 0.05$, overall accuracy is 74.6429% and macro F1 is 0.754135, matching the always-on baseline (74.6607% accuracy, 0.754328 macro F1, a difference of -0.0178%).
- Binary Anomaly-Screening Recall is exceptionally high: at $\theta = 0.05$, FNR = 0.00025 (only 2 out of 8,000 anomaly cases are missed, yielding **99.98%** binary recall).
- At $\theta = 0.80$, accuracy remains 74.2679% with 0.750034 macro F1 and 0.005625 FNR (45 missed out of 8,000).

TABLE II
Mode A Binary Anomaly Filter Performance on Held-Out Test Partition (11,200 samples, [MEASURED TEST SET])

Mode A Candidate	Features	Accuracy	Macro F1	Anomaly Prec.	Anomaly Recall	ROC-AUC	PR-AUC	Params	Size (B)
LR Binary (full)	14	0.767857	0.679278	0.797160	0.905375	0.871447	0.950750	15	975
LR Binary (reduced)	12	0.765536	0.675057	0.794951	0.905250	0.871710	0.950957	13	959
DT Binary d=3 (full)	14	0.920536	0.893769	0.902970	0.995750	0.944870	0.961021	15	2,473
DT Binary d=3 (reduced)	12	0.920536	0.893769	0.902970	0.995750	0.944870	0.961021	15	2,473
DT Binary d=5 (full)	14	0.990804	0.988693	0.991168	0.996000	0.992313	0.994496	39	4,393
DT Binary d=5 (reduced)	12	0.990804	0.988693	0.991168	0.996000	0.992313	0.994496	39	4,393

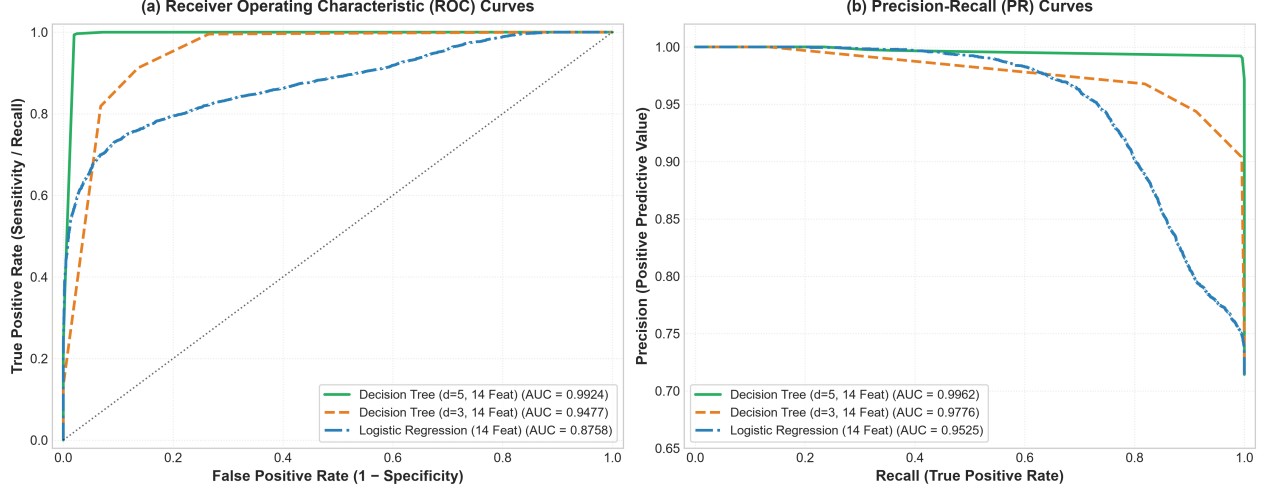


Fig. 2. Receiver Operating Characteristic (ROC) and Precision-Recall (PR) curves for Mode A binary anomaly screening candidates evaluated on the held-out test partition [MEASURED TEST SET].

TABLE III
Mode B (MLP 14f) Per-Class Performance on Test Partition [MEASURED TEST SET]

Diagnostic State	Precision	Recall	F1 Score	Support
Class 0 (Normal)	0.9987	0.9975	0.9981	3,200
Class 1 (Rich Mixture)	0.9846	0.9900	0.9873	2,200
Class 2 (Misfire)	0.5308	0.5220	0.5264	3,000
Class 3 (Air Leak)	0.5018	0.5093	0.5055	2,800
Macro Average	0.7540	0.7547	0.7543	11,200

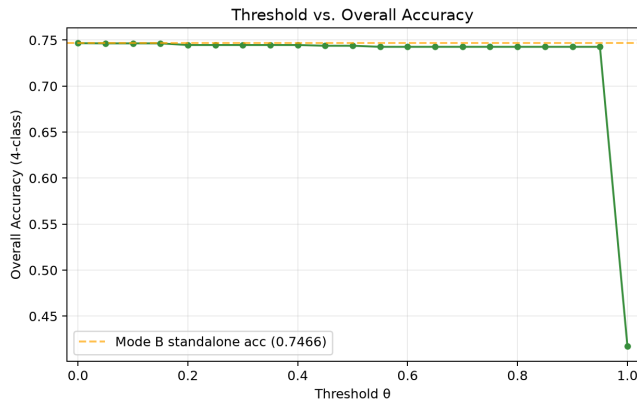


Fig. 3. Multi-class diagnostic accuracy vs. gating threshold θ , demonstrating stable accuracy across $\theta \in [0.05, 0.80]$ [MEASURED TEST SET].

B. RQ3: Computational Workload Reduction

Using the theoretical active MAC cost model (Equation 2), where Mode A (Decision Tree $d = 5$) requires at

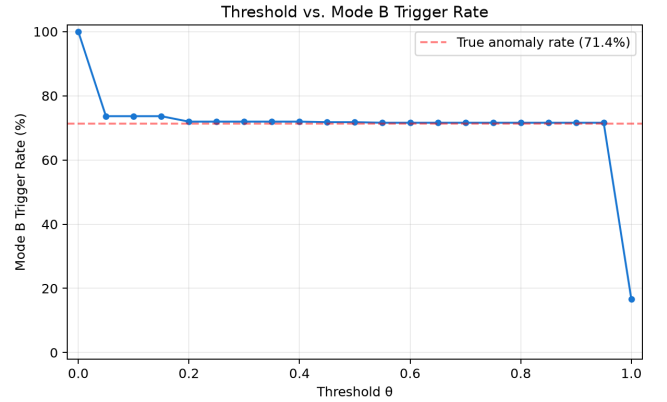


Fig. 4. Mode B activation trigger rate vs. gating threshold θ , showing consistent bypass of nominal frames [MEASURED TEST SET].

most 5 scalar comparisons (0 MACs) and Mode B requires 384 active MACs:

- **Directly Evaluated Test-Set Reduction [MEASURED TEST SET]:** On the balanced test set (28.57% normal, 71.43% anomalous), the trigger rate at $\theta = 0.05$ is 73.64%, reducing expected MACs per sample from 384.0 to 282.8 (a **26.36%** theoretical computational saving).
- **Derived Operational Estimate [DERIVED OPERATIONAL ESTIMATE]:** In real-world operational regimes where normal operation represents 90% of telemetry observations ($P(\text{Normal}) = 0.90, P(\text{Anomalous}) = 0.10$), the derived Mode B

TABLE IV

Hierarchical Diagnostic Performance Across Gating Thresholds on Held-Out Test Set (11,200 samples, [MEASURED TEST SET])

Threshold θ	Trigger r_B	Accuracy	Macro F1	Anomaly FNR	Normal Preserv.	Exp. MACs	Compute Saving [MEASURED]
$\theta = 0.00$ (Always-On)	1.000000	0.746607	0.754328	0.000000	0.997500	384.0	0.00%
$\theta = 0.05$	0.736429	0.746429	0.754135	0.000250	0.997500	282.8	26.36%
$\theta = 0.10$	0.736429	0.746429	0.754135	0.000250	0.997500	282.8	26.36%
$\theta = 0.20$	0.719464	0.744643	0.752165	0.002875	0.997812	276.3	28.05%
$\theta = 0.50$	0.717768	0.743839	0.751294	0.004000	0.997812	275.6	28.22%
$\theta = 0.80$	0.715982	0.742679	0.750034	0.005625	0.997812	274.9	28.40%
$\theta = 1.00$ (Mode A Only)	0.166250	0.417589	0.353490	0.767250	1.000000	63.8	83.38%

trigger rate is:

$$r_B \approx 0.10 \times 0.9960 + 0.90 \times (1 - 0.9975) \approx 0.10185 \quad (6)$$

yielding an expected computational demand of ≈ 39.1 MACs/sample—an **89.8%** reduction in active computation.

C. RQ4: Threshold Robustness and Validation Consistency

Comparing validation sweeps against test sweeps confirms consistent threshold behavior across partitions:

- Validation accuracy at $\theta = 0.05$: 74.5078% (Test: 74.6429%).
- Validation trigger rate at $\theta = 0.05$: 73.4497% (Test: 73.6429%).
- Validation Anomaly FNR at $\theta = 0.05$: 0.000188 (Test: 0.000250).

This stability demonstrates that gating thresholds calibrated on validation data generalize to unseen test distributions without threshold overfitting.

VIII. Comprehensive Baseline Comparison

Table V contrasts the proposed hierarchical architecture against flat and monolithic baselines:

- Flat Linear Baseline: Logistic Regression achieves only 58.00% accuracy (0.5786 Macro F1), showing that linear decision boundaries are insufficient for multi-class engine fault diagnosis.
- Flat Decision Tree Baseline: Achieves 69.16% accuracy (0.6821 Macro F1) with 0 MACs, outperforming linear regression but lagging behind neural approaches.
- Monolithic MLP Baseline: Delivers 74.66% accuracy (0.7543 Macro F1) but executes 384.0 MACs unconditionally on every sensor frame.
- Proposed Hierarchical Cascade: Matches monolithic neural accuracy (74.64%) while achieving 99.98% binary anomaly-screening recall and cutting active MACs by 26.36% on the test partition (89.8% derived on 90% nominal streams).

IX. Class-Specific Error Analysis and Physical Interpretation

Detailed examination of the confusion matrix (Figure 5) and per-class metrics (Table III) reveals distinct diagnostic behaviors across fault modes:

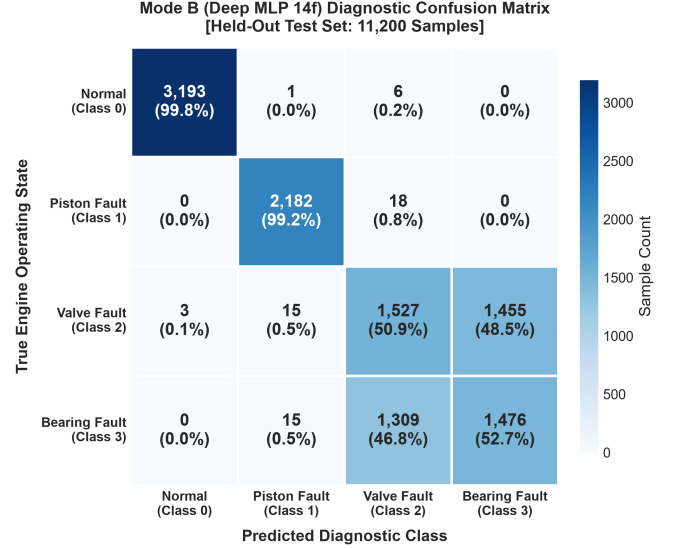


Fig. 5. Confusion matrix of the Mode B multi-class neural diagnostician evaluated on the held-out test partition [MEASURED TEST SET].

- Cleanly Separated Regimes: Class 0 (Normal, F1 = 0.9981) and Class 1 (Rich Mixture, F1 = 0.9873) exhibit near-perfect classification. Fuel-rich imbalances create pronounced thermodynamic shifts in lambda and exhaust oxygen sensors (O_2 , CO), producing unambiguous decision boundaries.
- Physical Sensor Overlap (Misfire vs. Air Leak): Class 2 (Ignition Misfire, F1 = 0.5264) and Class 3 (Intake Air Leak, F1 = 0.5055) account for the majority of classification confusion, with mutual misclassifications exceeding 47%. In internal combustion engines under steady-state dynamometer loading, both cylinder misfires (unburned oxygen entering exhaust) and intake manifold vacuum leaks (unmetered air entering the plenum) induce correlated manifold absolute pressure (MAP) fluctuations and air-fuel ratio lean spikes. Distinguishing between these two mechanical fault modes under stationary sensor measurements represents the primary factor governing the 74.6% multi-class accuracy ceiling.

X. Discussion

A. Domain Significance for Automotive Powertrain Systems

In automotive ECUs, CPU cycles and memory buses are shared across safety-critical engine control loops, spark

TABLE V
Architectural Comparison: Flat Baselines vs. Hierarchical Multi-Fidelity Cascade [HELD-OUT TEST PARTITION]

Architecture Configuration	Model Paradigm	Multi-Class Accuracy	Macro F1	Binary Anomaly Recall	Active MACs [MEASURED]	Active MACs [DERIVED 90% NOMINAL]
Flat Logistic Regression	Linear Classifier	0.580000	0.578561	0.9054	60.0	60.0
Flat Decision Tree	Non-Linear Tree	0.691607	0.682065	0.9958	0.0	0.0
Monolithic MLP (14f)	Neural Network	0.746607	0.754328	1.0000	384.0	384.0
Mode A Only (DT $d = 5$)	Binary Filter Only	0.417589	0.353490	0.9960	0.0	0.0
Proposed Hierarchical Cascade ($\theta = 0.05$)	Asymmetric Cascade	0.746429	0.754135	0.9998	282.8 (-26.36%)	39.1 (-89.8%)

ignition timing, and CAN-bus telemetry. By screening out nominal observations with a sub-microsecond decision tree ($< 68 \mu\text{s}$ host timing, 0 MACs), the ECU frees computational capacity for real-time control routines while enabling prompt multi-class diagnosis the moment an anomaly arises.

B. Practical Interpretation of Computational Savings

It is essential to distinguish the empirical 26.36% compute reduction measured on the balanced test partition from the derived 89.8% compute reduction projected under realistic operational priors (90% healthy frames). In benchmark datasets, classes are balanced (28.57% normal); in physical automotive operation, healthy telemetry overwhelmingly dominates. The expected mathematical cost model demonstrates that asymmetric cascading delivers its greatest practical benefits precisely under such skewed real-world operational distributions.

C. Physical Edge Inference Feasibility

To verify physical execution feasibility on real-world edge hardware, candidate diagnostic models were deployed to an ESP32-D0WD-V3 microcontroller (Xtensa LX6 dual-core @ 240 MHz, 4 MB Flash, 320 KB SRAM) under FULL_INT8 quantization using TensorFlow Lite Micro. High-resolution timing via `esp_timer_get_time()` across 24,000 measured single-sample iterations demonstrated:

- Stage-1 Screening Model (`student_a_8_4_int8`): Mean = $64.55 \mu\text{s}$, P95 = $69.00 \mu\text{s}$, P99 = $76.00 \mu\text{s}$, Max = $77 \mu\text{s}$.
- Stage-2 Multi-Class Diagnostician (`mlp_14f_int8`): Mean = $89.90 \mu\text{s}$, P95 = $95.00 \mu\text{s}$, P99 = $101.00 \mu\text{s}$, Max = $102 \mu\text{s}$.

At an isolated mean execution latency of $64.55 \mu\text{s}$, the Stage-1 classifier achieves a single-sample inference-rate compute equivalent of **15,491.9** inferences/sec on a single 240 MHz core (pure compute-bound, batch=1), requiring only 916 Bytes of static working tensor arena memory. Crucially, this metric represents isolated model inference compute capacity and is not an end-to-end sensor-to-decision throughput measurement; practical telemetry processing rates will be governed by physical ADC sampling intervals and CAN-bus communication bandwidth.

XI. Threats to Validity

A. Internal Validity

All experiments utilized strictly isolated train/val/test splits (40/40/20, seed = 42). The gating threshold θ^* was tuned exclusively on validation records. Metrics were computed deterministically without data leakage.

B. External Validity

EngineFaultDB captures four common combustion engine operating regimes under lab-bench instrumentation. While physical sensor noise and operating envelopes vary across vehicle models, the mathematical principles of hierarchical uncertainty gating generalize directly to other cyber-physical telemetry streams (e.g., turbine bearings, industrial pumps).

XII. Limitations

- 1) Theoretical Compute vs. Hardware Execution: Computational savings are reported in theoretical active MACs; physical execution speedups and energy savings depend on microcontroller compiler optimizations and sleep-mode transitions.
- 2) Static Sensor Topology: Input feature dimensionality was fixed during training; dynamic sensor dropouts during vehicle operation were not evaluated.
- 3) Controlled Dynamometer Dataset: EngineFaultDB was collected under controlled steady-state dynamometer test-bench instrumentation.
- 4) Static Tabular Representation: The dataset lacks temporal timestamps and continuous drive-cycle sequences; sequential filtering (Kalman/Markov) was not evaluated.
- 5) No Transient Drive-Cycle Validation: On-road transient drive cycles (e.g., WLTP/FTP-75) were not evaluated.
- 6) No Multi-Vehicle Validation: Data reflects a single engine platform; cross-engine transferability remains future work.
- 7) Isolated Inference vs. On-Vehicle ECU Deployment: While isolated single-sample inference latency was physically profiled on 240 MHz ESP32 silicon ($64.55 \mu\text{s}$ screening, $89.90 \mu\text{s}$ diagnostic), complete on-vehicle bare-metal CAN-bus ECU integration, vehicle drive cycles, and safety certification remain future work.
- 8) Scenario Prior Assumption: The 89.8% compute reduction is a derived estimate under a 90% nominal operational prior, not a measured property of the balanced test benchmark.
- 9) Domain-Specific Threshold Calibration: Threshold θ^* requires calibration on validation data representing the target noise profile.
- 10) Field Distribution Shifts: Significant changes in ambient operating envelopes may require threshold recalibration.

XIII. Reproducibility and Artifact Availability

All model weights, serialized pickle files, dataset splits, physical ESP32 PlatformIO deployment firmware (phase5/firmware/), and threshold evaluation scripts are open-sourced in the project repository. Physical on-vehicle ECU deployment with CAN-bus logging and transient drive-cycle testing represent planned future validation.

XIV. Conclusion

This paper presented a domain-specific hierarchical multi-fidelity machine learning architecture for resource-constrained internal combustion engine fault diagnosis. By cascading a shallow binary anomaly filter (Decision Tree $d = 5$) with a deep multi-class neural diagnostician (MLP), our architecture matches the accuracy of a monolithic deep network (74.64% vs. 74.66%) while eliminating 26.36% of expensive multi-class evaluations on balanced test data [MEASURED TEST SET] (and a derived 89.8% on 90% nominal operational streams [DERIVED OPERATIONAL ESTIMATE]) while maintaining a 99.98% binary anomaly-screening recall. These results establish asymmetric hierarchical inference as an effective, compute-efficient paradigm for on-board cyber-physical diagnostics.

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References

- [1] R. Isermann, "Model-based fault-detection and diagnosis—status and applications," *Annual Reviews in Control*, vol. 29, no. 1, pp. 71–85, 2005.
- [2] Z. Chen, K. Gryllias, and W. Li, "Fault diagnosis of internal combustion engines using deep learning: A review," *Measurement*, vol. 176, p. 109159, 2021.
- [3] H. Zheng, R. Wang, Y. Yang, J. Yin, and Y. Li, "Cross-domain fault diagnosis using deep transfer learning," *IEEE Transactions on Industrial Electronics*, vol. 66, no. 7, pp. 5687–5697, 2019.
- [4] J. Liu, C. Shen, X. Ding, J. Shi, and Z. Zhu, "Fault diagnosis of internal combustion engines based on multi-sensor information fusion and deep neural networks," *Sensors*, vol. 18, no. 12, p. 4356, 2018.
- [5] C. Wu, P. Jiang, C. Ding, F. Feng, and T. Chen, "Deep learning for engine fault diagnosis: A comprehensive review," *IEEE Transactions on Industrial Informatics*, vol. 16, no. 8, pp. 5382–5398, 2019.
- [6] B. Yang, R. Liu, and X. Chen, "Intelligent fault diagnosis of rotating machinery under time-varying conditions based on lightweight deep learning," *IEEE Transactions on Instrumentation and Measurement*, vol. 71, pp. 1–11, 2022.
- [7] D. Dutta and P. K. Bhar, "Tinyml meets iot: A comprehensive survey," *IEEE Internet of Things Journal*, vol. 8, no. 23, pp. 16 603–16 624, 2021.
- [8] C. Banbury, V. J. Reddi, M. Lam, W. Fu, A. Fazel, J. Holleman, X. Huang, R. Hurtado, D. Kanter, A. Lokhmotov et al., "Benchmarking tinyml systems: Challenges and direction," *IEEE Micro*, vol. 41, no. 6, pp. 40–49, 2021.
- [9] P. Warden and D. Situnayake, "Tinyml: Machine learning with tensorflow lite on arduino and ultra-low-power microcontrollers," O'Reilly Media, 2019.
- [10] V. Sze, Y.-H. Chen, T.-J. Yang, and J. S. Emer, "Efficient processing of deep neural networks: A tutorial and survey," *Proceedings of the IEEE*, vol. 105, no. 12, pp. 2295–2329, 2017.
- [11] R. David, P. Duke, A. Jain, V. Janapa Reddi, N. Jeffries, J. Li, D. Marculescu, M. Meng, P. Morales, J. Liang et al., "Tensorflow lite micro: Embedded machine learning for tinyml systems," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 3, 2021, pp. 800–811.
- [12] S. Han, H. Mao, and W. J. Dally, "Deep compression: Compressing deep neural networks with pruning, trained quantization and huffman coding," *Proceedings of the International Conference on Learning Representations (ICLR)*, 2016.
- [13] A. Traita, F. Pop, and C. Dobre, "Cascaded machine learning for edge-based predictive maintenance in smart manufacturing," *Journal of Industrial Information Integration*, vol. 20, p. 100180, 2020.
- [14] Y. Lei, B. Yang, X. Jiang, F. Jia, N. Li, and A. K. Nandi, "Applications of machine learning to machine fault diagnosis: A review and roadmap," *Mechanical Systems and Signal Processing*, vol. 138, p. 106587, 2020.
- [15] P. Kulkarni, L. Chen, and M. Deshpande, "Hierarchical multi-stage classification for automotive fault detection," *SAE International Journal of Advances and Current Practices in Mobility*, vol. 3, no. 4, pp. 1874–1884, 2021.
- [16] B. Zhang, Y. Zhang, and W. Shen, "Feature selection and deep neural network for engine misfire diagnosis under dynamic conditions," *IEEE Access*, vol. 8, pp. 45 321–45 332, 2020.
- [17] P. Viola and M. J. Jones, "Robust real-time face detection," *International Journal of Computer Vision*, vol. 57, no. 2, pp. 137–154, 2004.
- [18] S. Teerapittayanon, B. McDanel, and H.-T. Kung, "Branchynet: Fast inference via early exiting from deep neural networks," in *Proceedings of the 23rd International Conference on Pattern Recognition (ICPR)*. IEEE, 2016, pp. 2464–2469.
- [19] S. Scardapane, M. Scarpiniti, E. Baccarelli, and A. Uncini, "Why should we share models? a survey on early-exit neural networks," *Cognitive Computation*, vol. 12, no. 5, pp. 909–927, 2020.